FINITE-FIELD PARITY TREE SHIFTS: EXACT BOUNDARY COMPLEXITY, IID RAYS, AND COORDINATE-DELETION RECONSTRUCTION

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Abstract. Let q be a prime power, let $d \geq 2$, and fix nonzero coefficients $c_1, \dots, c_d \in \mathbb{F}_q$. We study labelings of the full rooted d-ary tree that satisfy the linear parity rule

$$x_w = \sum_{j=1}^{d} c_j x_{wj}$$

at every vertex. Every height-h block is parametrized uniquely by its d^h terminal labels, so the exact block count is q^{d^h}. This gives a complete normalization ledger: boundary-normalized complexity is $\log q$, site-normalized complexity tends to $(d-1)\log q/d$, and double-logarithmic growth is $\log d$.

Uniform terminal data define compatible uniform block laws and hence a shift-invariant probability measure. This measure is a homogeneous block-Markov chain with an affine-hyperplane offspring kernel. Although siblings are jointly constrained, the labels on every deterministic ray are iid uniform on $\mathbb{F}_q$. In the opposite direction, the full level at depth h reconstructs the root through an explicit nonzero linear functional. Every proper subset of that level is independent of the root: its mutual information with the root is exactly zero, while the complete level has mutual information $\log q$. Thus an iid process on each ray coexists with an exact coordinate-deletion threshold on every complete level. The tree-shift, entropy-normalization, joint-offspring, reconstruction, and secret-sharing frameworks are assigned to their prior owners; the residual contribution is this finite-field formula package, with no absolute priority claim.

1. Introduction

Local constraints on a rooted tree can be invisible along every individual lineage and still be perfectly visible across a complete generation. The example in this note makes that separation exact. Labels take values in a finite field, and every parent is a fixed nondegenerate linear combination of its children. A single ray therefore sees a new uniform field element at every step, while the exponentially wide boundary carries one field symbol of purely collective information about the root.

Tree shifts of finite type and their automata-theoretic foundations are due to Aubrun and Béal [1]. Ban and Chang developed the nonlinear-recurrence approach to finite-type tree-shift complexity and used the double-logarithmic entropy normalization [2]. Petersen and Salama introduced and established the site-normalized tree-shift entropy used below [7]. These frameworks are prior. We keep both normalizations visible because the same exact block count produces different constants under them.

Tree-indexed Markov processes form another established line, including the work of Benjamini and Peres [3]. In the binary case, a parent-to-pair kernel allowing conditional dependence between siblings is already the bifurcating Markov-chain framework used by Guyon [6]. Souissi later introduced block Markov chains on rooted trees [9]. Our children are not conditionally independent: given their parent, they are uniform on one affine hyperplane. We use the displayed joint-offspring factorization and claim no new Markov framework.

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Root reconstruction from noisy independent-edge broadcasts is likewise an established tree problem [5]. The law below is instead noiseless and sibling-coupled, and our result is a finite-depth statement about deleting coordinates from a complete level. We claim no general broadcasting or reconstruction theorem.

The residual results are four exact statements.

(1) Restriction to the terminal level is a bijection, giving $|\mathcal{B}_h| = q^{d^h}$ and an explicit root functional.
(2) Boundary, volume, and double-log normalizations are all evaluated from this one formula, without fitting a nonlinear recurrence.
(3) The compatible uniform law factors through a homogeneous joint offspring kernel, and every deterministic ray is iid uniform.
(4) The root has zero mutual information with every proper subset of a complete level but is determined by the complete level.

All four statements hold over every finite field, not only over prime fields. The accompanying control program performs exact enumeration over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4$, and $\mathbb{F}_5$, together with modular row reduction over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5$, and $\mathbb{F}_7$. The single $\mathbb{F}_4$ lane is an extension-field regression test; the all-prime-power scope comes from the proofs, not from experimental extrapolation.

Internal collision firewall. Two earlier internal systems require an explicit separation. P49 concerns complete cyclic *hom* tree-shifts defined by parent–child adjacency, transient phase allocation, and Hausdorff dimension in a boundary-weighted metric. The present shift instead has a star constraint coupling all siblings to their parent; it has no transient feeder, phase optimizer, graph homomorphism constraint, or Hausdorff-dimension claim. P77 concerns a one-dimensional countable orbit closure of fixed digit-weight automatic sets, its Cantor–Bendixson tower, and cellular-automaton endomorphisms. The present system is an uncountable free-semigroup tree shift and uses neither automatic digit support nor orbit-closure topology. Accordingly, no claim or proof from either internal paper is reused.

2. The parity tree shift

Fix an integer $d \geq 2$ and write $D = \{1, \dots, d\}$. The vertex set of the full rooted ordered d-ary tree is the free monoid

$$\mathcal{T}_d = D^* = \bigcup_{h \geq 0} D^h,$$

whose root is the empty word ε. Put

$$\mathsf{L}_h = D^h, \qquad \mathsf{V}_h = \bigcup_{r=0}^{h} \mathsf{L}_r.$$

Thus $|\mathsf{L}_h| = d^h$ and

$$|\mathsf{V}_h| = 1 + d + \cdots + d^h = \frac{d^{h+1} - 1}{d - 1}. \tag{1}$$

For a labeling $x \in \mathbb{F}_q^{\mathcal{T}_d}$ and $j \in D$, define the subtree shift by $(\sigma_j x)_w = x_{jw}$.

Let q be a prime power and let

$$\boldsymbol{c} = (c_1, \dots, c_d) \in (\mathbb{F}_q^\times)^d. \tag{2}$$

Definition 2.1 (Parity tree shift)**.** The finite-field parity tree shift is

$$X_{q,d,\boldsymbol{c}} = \left\{ x \in \mathbb{F}_q^{\mathcal{T}_d} : x_w = \sum_{j=1}^{d} c_j x_{wj} \text{ for every } w \in \mathcal{T}_d \right\}. \tag{3}$$

Let $\mathcal{B}_h(X_{q,d,\boldsymbol{c}})$ be the set of its restrictions to V_h.

The rule is a finite list of allowed labeled stars, so $X_{q,d,\boldsymbol{c}}$ is a tree shift of finite type. It is preserved by each σ_j. When $q = 2$ and $c_1 = \cdots = c_d = 1$, the rule says that the parent is the parity, or xor, of its children. The terminology "parity" will also be used for its finite-field linear extension.

For $u = u_1 \cdots u_h \in \mathsf{L}_h$, set

$$c_u = \prod_{r=1}^{h} c_{u_r}, \qquad c_\varepsilon = 1. \tag{4}$$

Every c_u is nonzero by (2).

Theorem 2.2 (Exact boundary parametrization)**.** *For every $h \geq 0$, restriction to the terminal level is a bijection*

$$\mathcal{B}_h(X_{q,d,\boldsymbol{c}}) \longrightarrow \mathbb{F}_q^{\mathsf{L}_h}, \qquad b \longmapsto b|_{\mathsf{L}_h}. \tag{5}$$

For the inverse block, every vertex is reconstructed upward. In particular,

$$b_\varepsilon = \sum_{u \in \mathsf{L}_h} c_u b_u. \tag{6}$$

Consequently

$$|\mathcal{B}_h(X_{q,d,\boldsymbol{c}})| = q^{d^h}, \tag{7}$$

and exactly q^{d^h-1} of these blocks have any prescribed root label $a \in \mathbb{F}_q$.

Proof. Starting with arbitrary labels on L_h, apply (3) successively at levels $h-1, h-2, \ldots, 0$. This constructs one legal block and shows both existence and uniqueness on V_h. The block extends to an infinite tree: for each terminal vertex, choose $d-1$ child labels arbitrarily and solve for the last one using $c_d \neq 0$; repeat at every new terminal level. Thus every upward-constructed block belongs to $\mathcal{B}_h(X_{q,d,\boldsymbol{c}})$.

Iterating the local equation gives (6). More explicitly, substituting the children of the root once produces coefficients c_j; each additional substitution multiplies the coefficient along the new edge, yielding c_u after h levels. The bijection now gives (7). Finally, the right side of (6) is a nonzero linear functional on the d^h-dimensional space $\mathbb{F}_q^{\mathsf{L}_h}$. Each fiber is an affine hyperplane of cardinality q^{d^h-1}. $\square$

The root-refined block count also closes the Ban–Chang nonlinear recurrence in one line. If $A_h(a)$ denotes the number of height-h blocks rooted at a, then $A_h(a)$ is independent of a; writing the common value as A_h gives

$$A_0 = 1, \qquad A_{h+1} = q^{d-1} A_h^d, \qquad A_h = q^{d^h-1}. \tag{8}$$

Indeed there are q^{d-1} child tuples with a prescribed parent, and the d rooted subblocks can then be selected independently. Equation (8) is a specialization of the prior nonlinear- recurrence framework, not a new recurrence formalism.

3. A normalization ledger

Because tree volumes grow exponentially with height, the word "entropy" has been used for more than one normalization. We therefore state the unambiguous finite-height quantity first:

$$C_h := \log |\mathcal{B}_h(X_{q,d,\boldsymbol{c}})| = d^h \log q. \tag{9}$$

All logarithms are natural.

TABLE 1. The same exact block count under three normalizations. The first row is retained as a boundary complexity rather than called a third standard entropy.

Normalization	Finite-height value	Limit
terminal level	C_h/d^h	$\log q$
all sites	$C_h/\lvert\mathsf{V}_h\rvert$	$(d-1)\log q/d$
double logarithm ($h \geq 1$)	$(\log C_h)/h$	$\log d$

Theorem 3.1 (Exact normalized complexity)**.** *For every $h \geq 0$, the boundary- and site-normalized complexities satisfy the first two identities below. For every $h \geq 1$, the double-logarithmic complexity satisfies the third:*

$$\frac{C_h}{|\mathsf{L}_h|} = \log q, \tag{10}$$

$$\frac{C_h}{|\mathsf{V}_h|} = \frac{(d-1)d^h}{d^{h+1}-1}\log q \longrightarrow \frac{d-1}{d}\log q, \tag{11}$$

$$\frac{\log C_h}{h} = \log d + \frac{\log\log q}{h} \longrightarrow \log d. \tag{12}$$

In particular, the Petersen–Salama site-normalized entropy is $(d-1)\log q/d$, while the Ban–Chang double-logarithmic entropy is $\log d$.

Proof. Insert (9), $|\mathsf{L}_h| = d^h$, and (1). The three displayed identities and their limits follow directly. □

The coefficient $\log q$ in $|\mathcal{B}_h| = \exp((\log q)d^h)$ records one free field symbol per terminal site. By contrast, the full tree shift on q symbols has one free symbol at every site and hence the larger site-normalized rate $\log q$. The parity rule removes precisely the internal-site degrees of freedom.

4. The compatible uniform block-Markov law

Let ν_h be the uniform probability on $\mathcal{B}_h(X_{q,d,\boldsymbol{c}})$. The next proposition shows that these finite laws define one infinite law without a boundary choice at infinity.

Proposition 4.1 (Compatibility and invariance)**.** *The restriction map from height $h+1$ to height h has constant fiber size*

$$q^{(d-1)d^h} = q^{d^{h+1}-d^h}. \tag{13}$$

Hence the laws $(\nu_h)_{h\geq 0}$ are projectively compatible and determine a probability measure $\mu_{q,d,\boldsymbol{c}}$ on $X_{q,d,\boldsymbol{c}}$. It satisfies

$$\mu_{q,d,\boldsymbol{c}}([b]) = q^{-d^h} \qquad (b \in \mathcal{B}_h), \tag{14}$$

and is invariant under every subtree shift σ_j.

Proof. Fix a legal block on V_h. At each of its d^h terminal vertices, a child tuple must satisfy one nonzero linear equation in d variables. It has q^{d-1} solutions. The choices at distinct terminal vertices are independent, proving (13). Uniform measures therefore push forward to uniform measures. The projective-limit measure exists because the alphabet is finite, and (14) follows from (7).

For shift invariance, fix a vertex v and a height h. The labels at the d^h descendants vu, $u \in \mathsf{L}_h$, are an iid uniform subcollection of the uniform level $|v| + h$. Upward reconstruction inside the subtree rooted at v therefore gives the uniform law on $\mathcal{B}_h$. Cylinder sets generate the Borel sigma-field, so every σ_j preserves $\mu_{q,d,\boldsymbol{c}}$. □

For $a \in \mathbb{F}_q$ and $y = (y_1, \ldots, y_d) \in \mathbb{F}_q^d$, define

$$K(a; y_1, \ldots, y_d) = q^{-(d-1)} \mathbf{1}\left\{ a = \sum_{j=1}^{d} c_j y_j \right\}. \tag{15}$$

This is a probability kernel from a parent to its ordered child block. For $h \geq 1$ and a labeling b of V_h,

$$\mu_{q,d,\boldsymbol{c}}([b]) = q^{-1} \prod_{w \in \mathsf{V}_{h-1}} K(b_w; b_{w1}, \ldots, b_{wd}), \tag{16}$$

with the convention that the right side is zero when a local rule fails. For $h = 0$, the factorization consists only of the root factor q^{-1}. For a legal block, the exponent is $1 + (d-1)|\mathsf{V}_{h-1}| = d^h$, so (16) agrees with (14). Thus the measure is a homogeneous block Markov chain in the joint-offspring sense of [9].

Lemma 4.2 (Local proper-subset uniformity). *Fix a parent value $a \in \mathbb{F}_q$. Under the kernel $K(a; \cdot)$, every proper subset of the d child coordinates is iid uniform on the corresponding power of $\mathbb{F}_q$, and its law does not depend on a.*

Proof. Fix $s < d$ child coordinates and prescribe their values. At least one child coordinate remains free, and its coefficient is nonzero. The residual linear equation in $d - s$ variables has exactly q^{d-s-1} solutions. There are q^{d-1} offspring tuples in total. The probability of each prescribed s-tuple is therefore $q^{d-s-1}/q^{d-1} = q^{-s}$, independently of the parent value. □

The lemma shows why a standard independent-offspring branching description would be inaccurate: a complete sibling block lies on a codimension-one hyperplane. Every proper sibling subblock nevertheless looks exactly independent.

Theorem 4.3 (Every ray is iid). *Let*

$$w_0 = \varepsilon, \qquad w_{r+1} = w_r j_{r+1} \quad (r \geq 0)$$

be any deterministic ray. Under $\mu_{q,d,\boldsymbol{c}}$, the process $(X_{w_r})_{r \geq 0}$ is iid with common law $\mathrm{Unif}(\mathbb{F}_q)$. *Equivalently, for all $h \geq 0$ and $a_0, \ldots, a_h \in \mathbb{F}_q$,*

$$\mu_{q,d,\boldsymbol{c}}\{X_{w_0} = a_0, \ldots, X_{w_h} = a_h\} = q^{-(h+1)}. \tag{17}$$

Proof. The root is uniform by (14) with $h = 0$. By (16), conditional on the labels through one generation, the ordered offspring block at each exposed vertex has kernel K. Lemma 4.2 with a single selected child says that the next label on the ray is uniform and its conditional law does not depend on the parent or on the earlier ray labels. Multiplying these $h+1$ conditional probabilities proves (17). □

Ray iid should not be confused with independence of the whole tree. The next section gives the strongest possible obstruction: one complete level determines the root exactly.

5. Exact coordinate-deletion reconstruction

Let $R = X_\varepsilon$ and let $Z_h = (X_u)_{u \in \mathsf{L}_h}$. For a subset $A \subseteq \mathsf{L}_h$, write $Z_A = (X_u)_{u \in A}$. Shannon entropy and mutual information are computed with natural logarithms.

For a fixed depth, the information condition "all coordinates recover, while every proper coordinate subset reveals nothing" is the perfect (n, n) threshold criterion from classical secret sharing, introduced independently by Blakley and Shamir [4, 8]. We claim neither that access structure nor its additive finite-field mechanism. The calculation below identifies its simultaneous realization at every depth by one shift-invariant tree law.

Theorem 5.1 (Exact reconstruction threshold). *For every $h \geq 1$, the complete level is an iid uniform vector in $\mathbb{F}_q^{d^h}$ and*

$$R = \sum_{u \in \mathsf{L}_h} c_u X_u \quad \textit{almost surely, and indeed pointwise on } X_{q,d,\boldsymbol{c}}. \tag{18}$$

For every proper subset $A \subsetneq \mathsf{L}_h$,

$$\mu(R = a, Z_A = z) = q^{-(|A|+1)} \quad (a \in \mathbb{F}_q,\ z \in \mathbb{F}_q^A). \tag{19}$$

Consequently

$$\mathrm{H}(R \mid Z_A) = \log q, \qquad \mathrm{I}(R; Z_A) = 0, \tag{20}$$

$$\mathrm{H}(R \mid Z_h) = 0, \qquad \mathrm{I}(R; Z_h) = \log q. \tag{21}$$

Proof. The terminal-level bijection and the uniform block law show that Z_h is uniform on $\mathbb{F}_q^{\mathsf{L}_h}$, hence its coordinates are iid uniform. Equation (18) is (6).

Fix $A \subsetneq \mathsf{L}_h$ and condition on $Z_A = z$. Choose one omitted vertex $v \in \mathsf{L}_h \setminus A$. All coefficients c_u are nonzero, so $c_v X_v$ is uniform on $\mathbb{F}_q$ and independent of the conditioned coordinates. Even after conditioning on every other omitted coordinate, the sum in (18) is therefore uniform. Removing that extra conditioning leaves $\mu(R = a \mid Z_A = z) = q^{-1}$. Since $\mu(Z_A = z) = q^{-|A|}$, this proves (19).

The root is uniform, so $\mathrm{H}(R) = \log q$. Independence gives (20); deterministic reconstruction gives (21). □

Thus deleting even one label from a complete level removes all information about the root, not merely some fraction of it. The phenomenon is high-order: every finite ray is a product process by Theorem 4.3, and the whole generation is also an unconditional product vector, yet one linear combination of all generation coordinates equals the root.

Remark 5.2 (Role of the assumptions). The condition $d \geq 2$ is needed for ray iid: with one child the local rule determines that child from its parent. Nonzero coefficients are needed for the all-proper-subsets statement. If $c_j = 0$, then at level one the root is already reconstructed after child j is omitted, so (20) fails. Boundary parametrization alone is less restrictive; the assumptions are frozen because the combined theorem package requires all of them.

6. Owner subtraction, controls, and scope

Table 2. Ownership and collision boundary. "Residual here" means the specialized calculation proved in this note, not a priority assertion.

Source or internal system	Owned framework	Residual here
Aubrun–Béal	finite-type and sofic tree shifts	one affine-star family
Ban–Chang	SNRE and double-log entropy	closed scalar count q^{d^h}
Petersen–Salama	site-normalized tree entropy	exact rate $(d-1)\log q/d$
Benjamini–Peres	tree-indexed Markov processes	no new general process
Guyon; Souissi	joint sibling kernels and block-Markov language	explicit d-ary affine kernel and iid rays
Evans et al.	tree broadcasting and root reconstruction	noiseless coordinate-deletion law
Blakley; Shamir	perfect threshold secret sharing	one law at every tree depth
Internal P49	hom-tree Hausdorff phase allocation	no parent–child graph or phase optimization
Internal P77	one-dimensional automatic towers	no automatic support or Cantor–Bendixson claim

The broad object, both entropy conventions, the joint-offspring and block-Markov language, noisy reconstruction, and the threshold access condition are therefore subtracted from the contribution. A bounded primary-source audit through 28 August 2026 did not locate the combined finite-field formula package stated here. That sentence records a search boundary only; it is not a claim of worldwide novelty. Public posting, submission, author contact, and priority language remain on **HOLD**.

The accompanying script `code/verify_parity_tree.py` uses only the Python standard library. Its two lanes are deliberately redundant.

(1) The enumeration lane constructs every block from every terminal word, checks local legality and injectivity, brute-forces all legal labelings in the smallest cases, measures restriction-fiber multiplicities, enumerates every deterministic ray, and compares complete joint tables for the root and every proper terminal subset in selected cases.
(2) The rank lane forms the local constraint matrix over $\mathbb{F}_p$, verifies its rank and nullity, constructs the full boundary-extension matrix, checks that it lies in the constraint kernel, and certifies that adjoining the root functional increases the rank of every tested proper observation matrix by one.

The recorded run covers 8,315 enumerated blocks, 811 exhaustive proper subsets, 78 constraint rows, 567 observation-rank certificates, and 19,764 assertions. These finite controls are not proofs of arbitrary q, d, h. The implementation has one exhaustive $\mathbb{F}_4$ lane, while its rank lane handles prime fields only; all-prime-power scope rests on the finite-field arguments in Theorems 2.2 and 5.1.

7. Conclusion

The parity tree shift has one free field coordinate per terminal vertex and no other finite-block degrees of freedom. That leaf parametrization gives the entire complexity ledger and a canonical compatible uniform measure. The same measure has two opposing exact faces: every lineage is iid, while every complete generation reconstructs the root and loses all root information after any coordinate is deleted. The coefficient of each terminal label is simply the product of the edge coefficients along its path, so nondegeneracy at one local star propagates to every depth.

The scope is intentionally narrow. We do not classify general linear tree shifts with several equations, spatially varying coefficients, missing coefficients, noisy parity checks, or nonlinear local rules. Those changes can alter rank density, compatibility, or the sharp reconstruction threshold. A natural next problem is to replace the single parity equation by a matrix-valued local code and determine which rank profiles preserve iid rays while producing a nontrivial hierarchy of boundary information.

References

[1] Nathalie Aubrun and Marie-Pierre Béal. Tree-shifts of finite type. *Theoretical Computer Science*, 459:16–25, 2012. doi: 10.1016/j.tcs.2012.07.020.

[2] Jung-Chao Ban and Chih-Hung Chang. Tree-shifts: The entropy of tree-shifts of finite type. *Nonlinearity*, 30(7):2785–2804, 2017. doi: 10.1088/1361-6544/aa72c0.

[3] Itai Benjamini and Yuval Peres. Markov chains indexed by trees. *The Annals of Probability*, 22(1): 219–243, 1994. doi: 10.1214/aop/1176988857.

[4] George Robert Blakley. Safeguarding cryptographic keys. In *Proceedings of the 1979 AFIPS National Computer Conference*, volume 48, pages 313–317. AFIPS Press, 1979. doi: 10.1109/AFIPS.1979.98.

[5] William Evans, Claire Kenyon, Yuval Peres, and Leonard J. Schulman. Broadcasting on trees and the Ising model. *The Annals of Applied Probability*, 10(2):410–433, 2000. doi: 10.1214/aoap/1019487349.

[6] Julien Guyon. Limit theorems for bifurcating markov chains: Application to the detection of cellular aging. *The Annals of Applied Probability*, 17(5–6):1538–1569, 2007. doi: 10.1214/105051607000000195.

[7] Karl Petersen and Ibrahim Salama. Tree shift topological entropy. *Theoretical Computer Science*, 743:64–71, 2018. doi: 10.1016/j.tcs.2018.05.034.

[8] Adi Shamir. How to share a secret. *Communications of the ACM*, 22(11):612–613, 1979. doi: 10.1145/359168.359176.

[9] Abdessatar Souissi. Block markov chains on trees. In *Infinite Dimensional Analysis, Quantum Probability and Applications*, pages 151–165. Springer International Publishing, 2022. ISBN 978-3-031-06170-7. doi: 10.1007/978-3-031-06170-7_8.